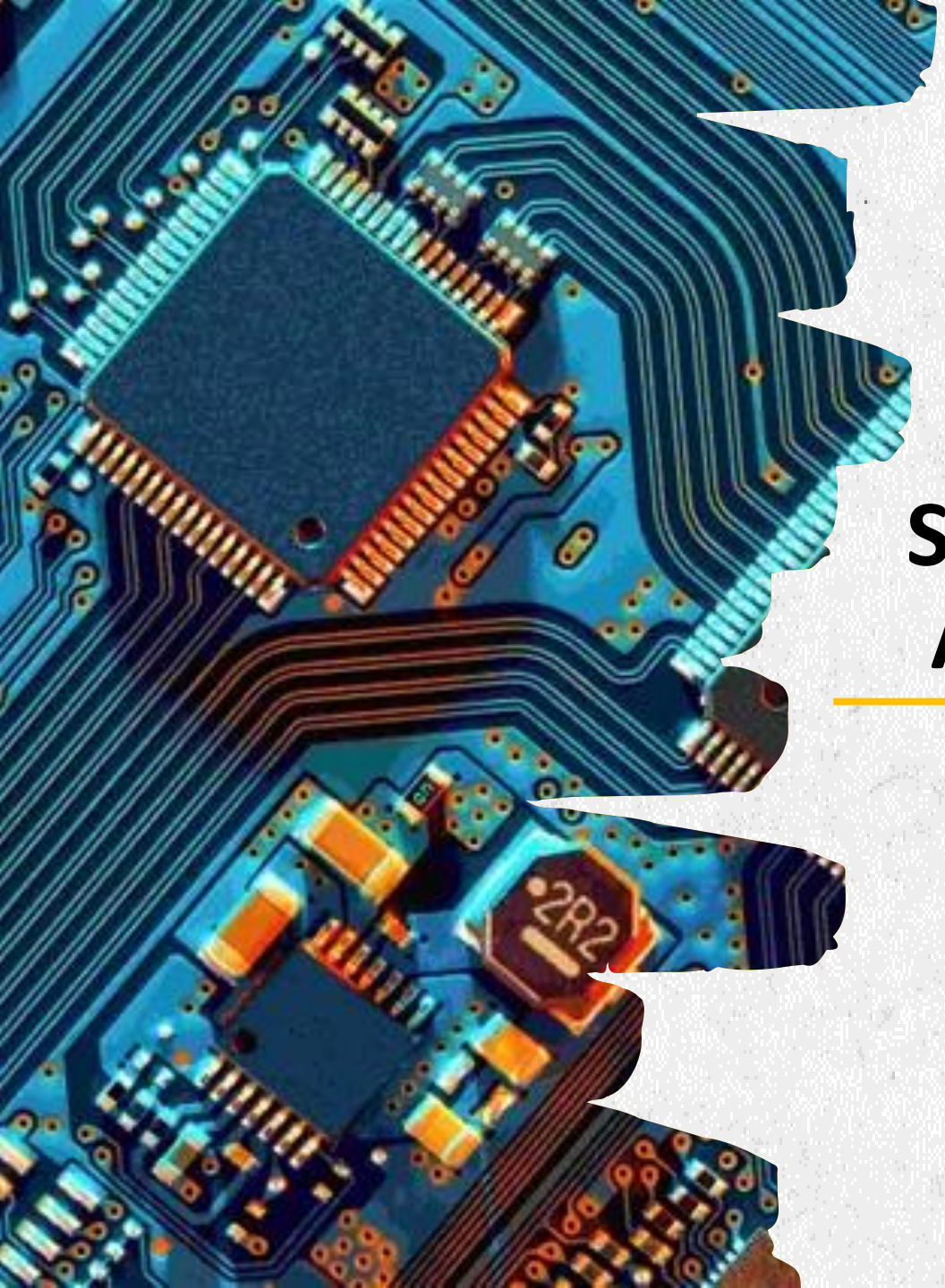
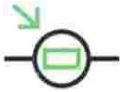
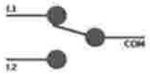
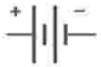


CHAPTER 2
SEMICONDUCTORS AND ELECTRONICS

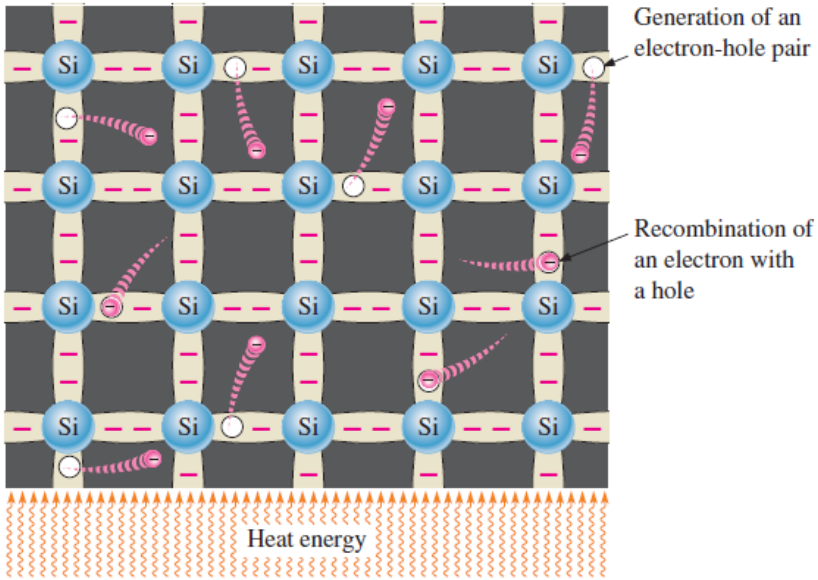
電路學

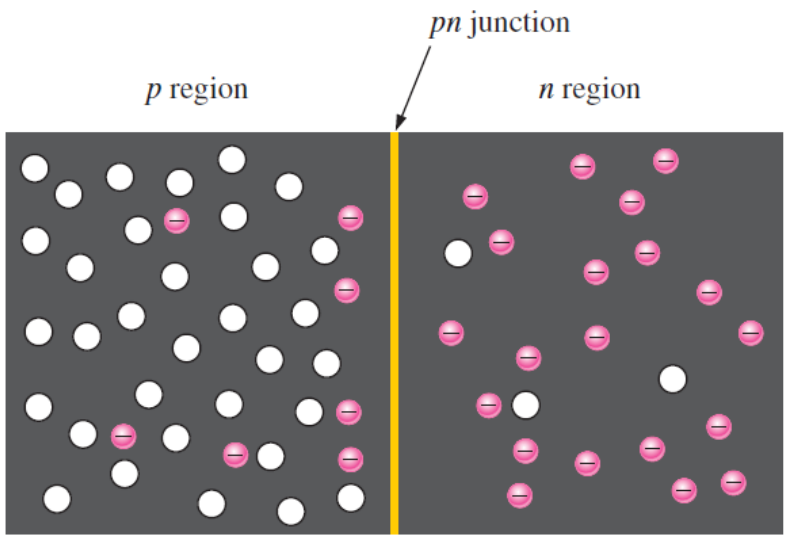
應用電子學

PASSIVE			ACTIVE		
Resistor			Transistor		
LDR			Diode		
Thermistor			LED		
Capacitor			Photodiode		
Inductor			Integrated Circuit		-
Switch			Operational Amplifier		
Variable Resistor			Seven Segment Display		
Transformer			Battery		

INTRODUCTION

- Most of *active electronic devices (主動元件)*, such as *diodes* (二極體), *transistors* (晶体管), *Op-Amp* (運算放大器), *and integrated circuits* (*ICs*、積體電路) are made of *semiconductive materials* (半導體材料).
- To understand how these devices work, you should have a basic knowledge their atomic structure (原子結構) and so-called the *pn* junction (*pn*接面).





PERIODIC TABLE OF ELEMENTS

Key: 29 Cu 63.55 — Atomic number, Symbol, Atomic weight

Metal | Nonmetal | Intermediate

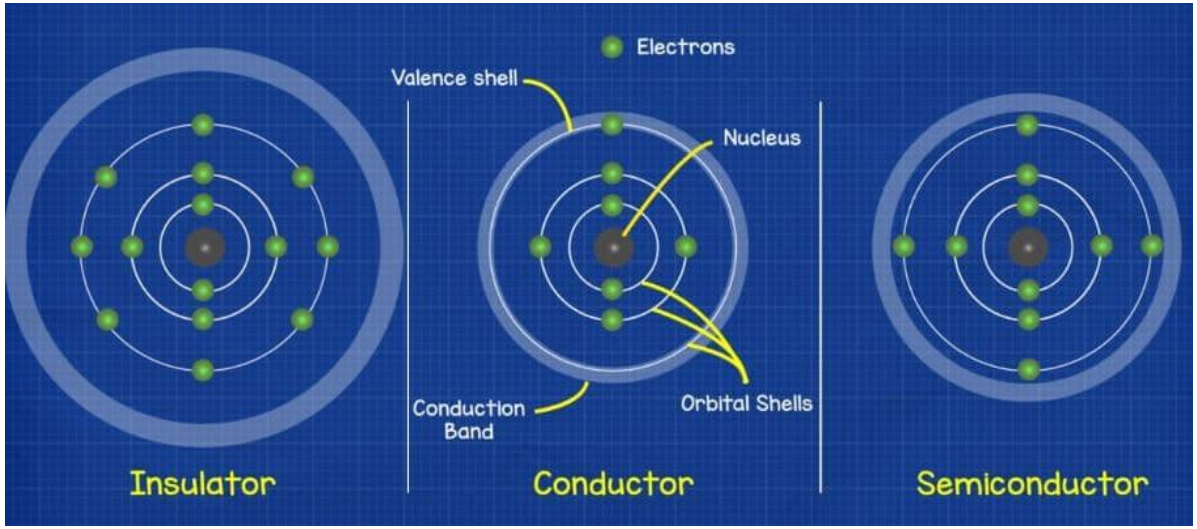
IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	O
1 H 1.0080																	2 He 4.0026
3 Li 6.941	4 Be 9.0122											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.064	17 Cl 35.453	18 Ar 39.948
19 K 39.098	20 Ca 40.08	21 Sc 44.956	22 Ti 47.87	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41	31 Ga 69.72	32 Ge 72.64	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.4	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.30
55 Cs 132.91	56 Ba 137.33	Rare earth series	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.2	76 Os 190.23	77 Ir 192.2	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.19	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	Acti-nide series	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)								

Rare earth series	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.35	63 Eu 151.96	64 Gd 157.25	65 Tb 158.92	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
Actinide series	89 Ac (227)	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

MATERIALS USED IN ELECTRONICS

- In terms of their *electrical properties (電路屬性)*, materials can be classified into three groups:
  1. **Conductors (電導體)** – a material that easily conducts electrical current. The best conductors are ***single-element metal materials***, such as Cu 銅, Ag 銀, Au 金, and Al 鋁.
  2. **Insulators (絕緣體) –** a material that does not conduct electrical current. Most good insulators are ***compounds of metal and nonmetal elements***, such as rubber, glass, plastics.
  3. **Semiconductor (半導體) –** a material that is ***between conductors and insulators in its ability to conduct electrical current***. The most common single-element semiconductors are Si 矽 and Ge 鍺.

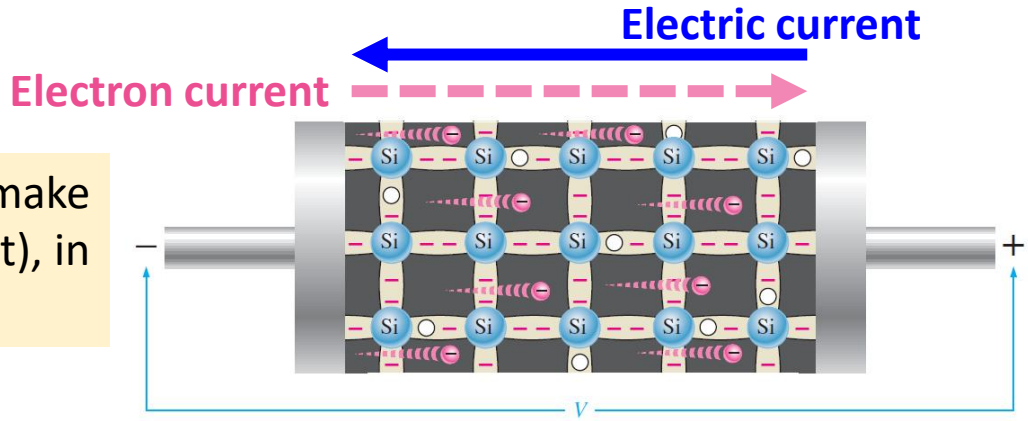
➢ What causes a material to become a *conductor, insulator, or semiconductor*?



Conductivity depends on *the electrons on the valence (outermost) shell 價層*.

1. ***Low conductivity*** – The valence shell is ***packed with electrons (mostly 6~8 electrons).*** The ***strong bond 鍵*** does not allow electrons 電子 to be free.

2. ***High conductivity*** – The valence shell has ***only a few electrons***. The ***bond 鍵is very weak***; electrons get free easily.

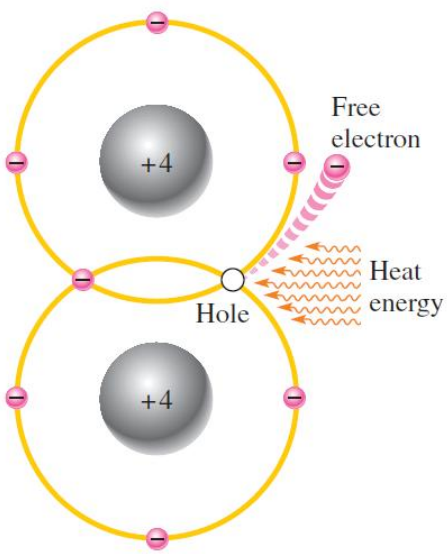


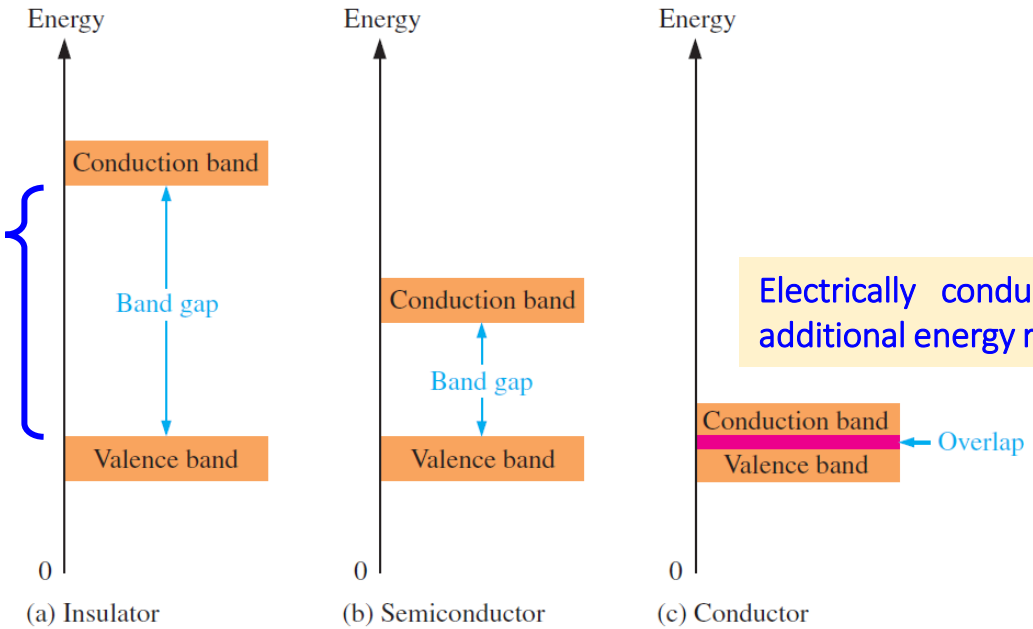
Once electrons are free to flow, it can make the current of energy (electricity current), in the reverse direction.

BAND GAP

- <u>Band gap 帶隙 (or energy gap 能隙)</u> is the additional energy required to free electrons from the valence shell.

Need large energy to free electrons (or make this material electrically conductive)





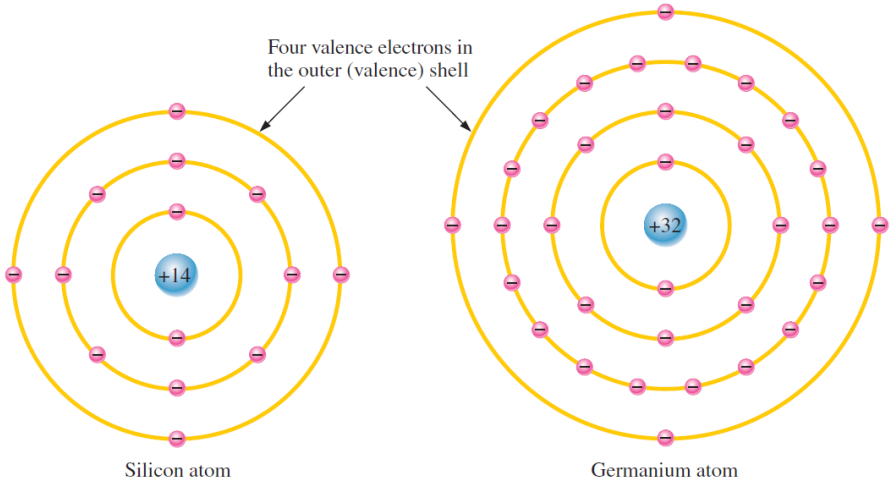
Electrically conductive with no additional energy required

Need some energy to free electrons (or make this material electrically conductive)

SILICON AND GERMANIUM

- Most of *semiconductor devices (半導體元件)* are made of either **Silicon (Si 矽) or Germanium (Ge 鍺)**, both of which have *4 valence electrons 價電子*.

- Germanium (Ge 鍺) requires a smaller additional energy to free electrons, being *too sensitive at high temp*.

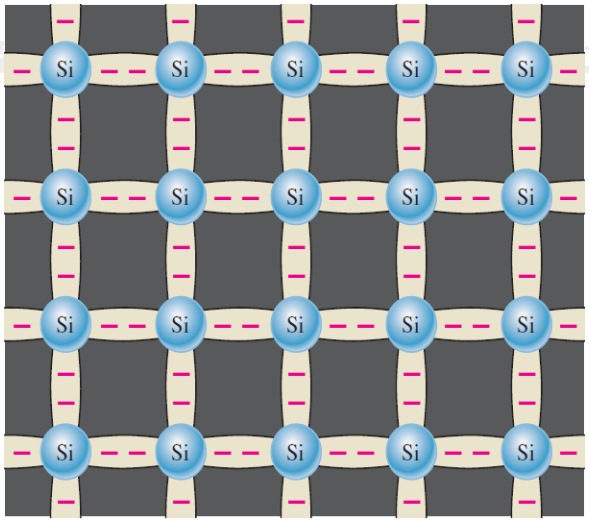


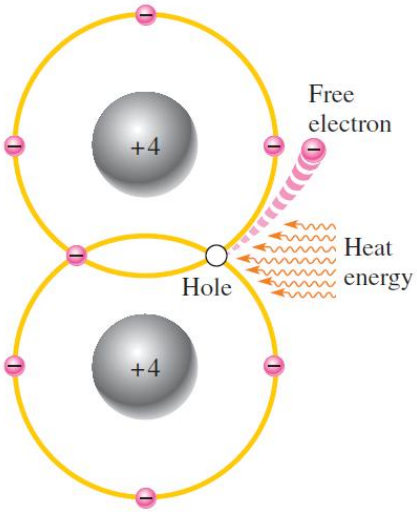
➢ This is why *silicon (Si 矽) is a more widely used semiconductive material*.

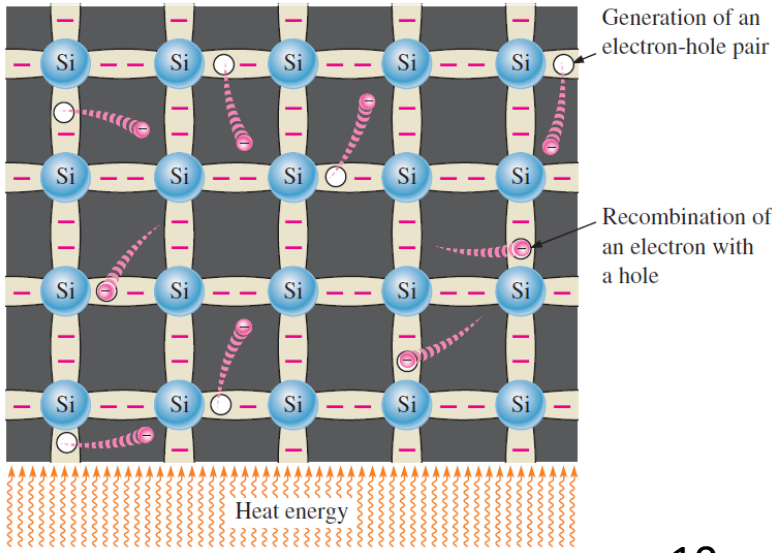
➢ *Silicon Valley 矽谷* is also titled due to this reason.

ELECTRONS AND HOLES

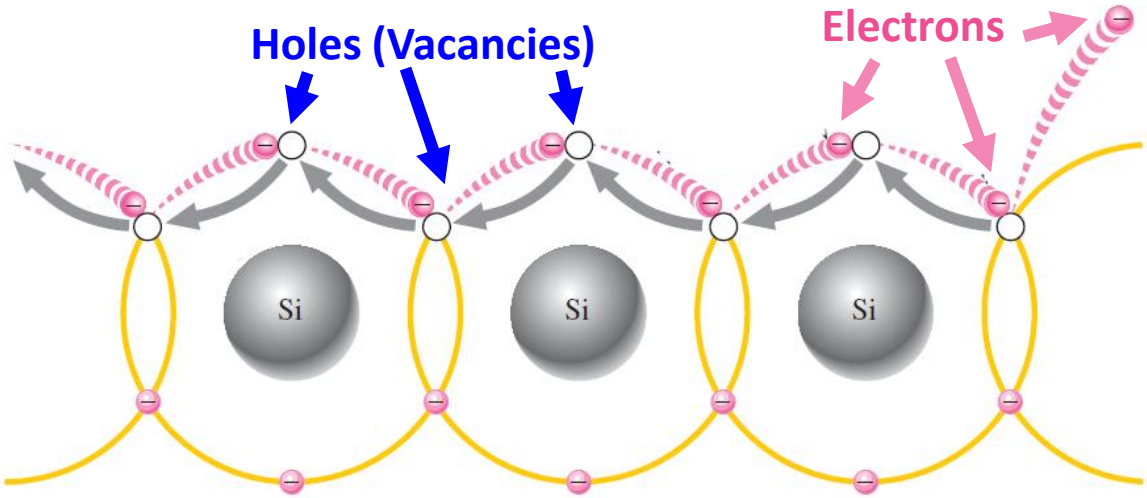
- Numerous silicon atoms connect and share electrons, forming the *crystal structure* 晶體結構.
- At room temp., some free electrons *randomly drift throughout the material*.
- While an *electron* jumps, a *vacancy* 空缺 is left within the structure.
- The vacancy is called a "*hole* 電子孔".

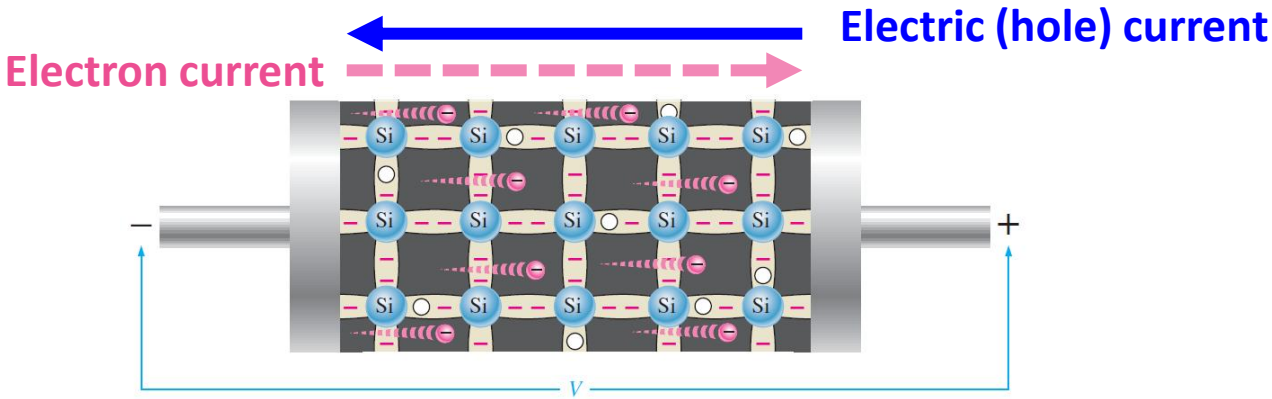






- When a voltage is applied across a piece of silicon, the free electrons are now easily move toward the positive end. It is called "*electron current*".
- Inversely, the same amount of holes move toward the negative end. This is called "*hole current*", which is the *electric current* (電流).

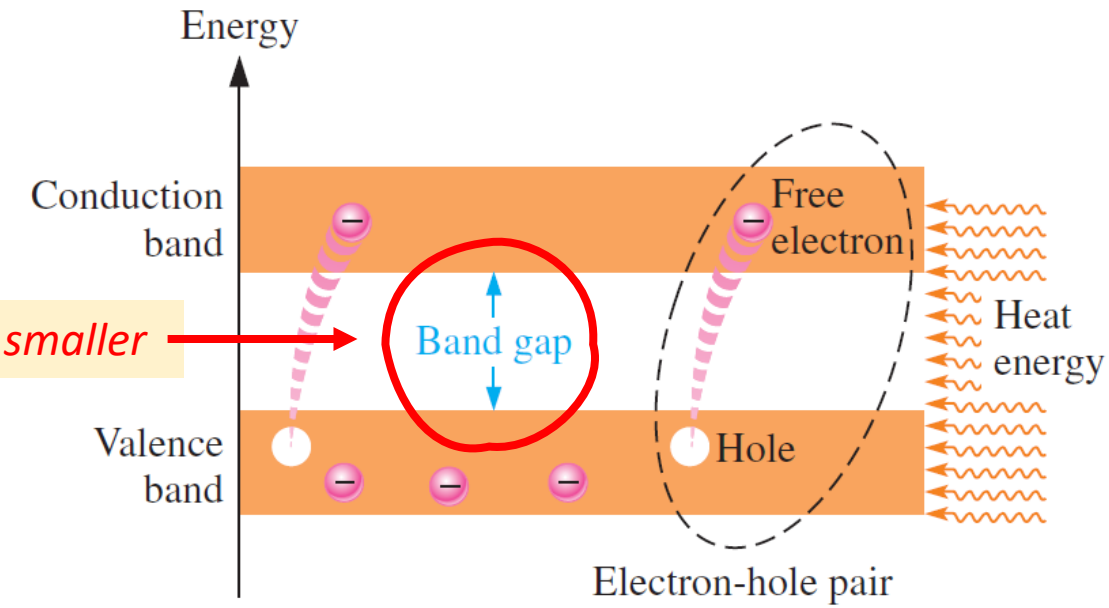




P-TYPE AND N-TYPE SEMICONDUCTORS

- The *pure semiconductive* material are generally *poor conductors*.
- To improve their conductivity, we can *add impurities (雜質) to the material*, that is adding other type of material. This process is called the "*doping 摻雜*" process, which has two categories: (1) ***n*-type** and (2) ***p*-type**.

***Doping** makes the band gap get smaller*

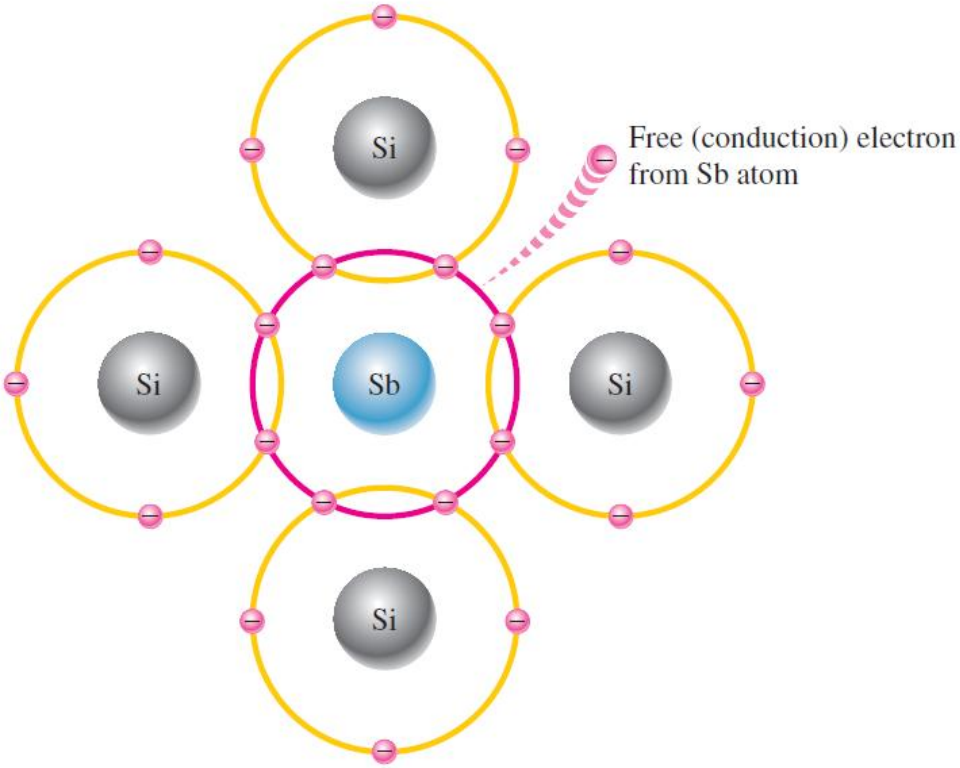


N-TYPE SEMICONDUCTORS

- *N-type semiconductors* are made by adding *5 valence electrons materials*, such as arsenic (As 砷), phosphorus (P 磷), bismuth (Bi 鉍), and antimony (Sb 銻), into the pure silicon or germanium.

- ***N-type semiconductors*** have:
  - more electrons (majority)
  - less hole (minority)

- The "***N***" stands for the negative charge 負電荷 on electron.

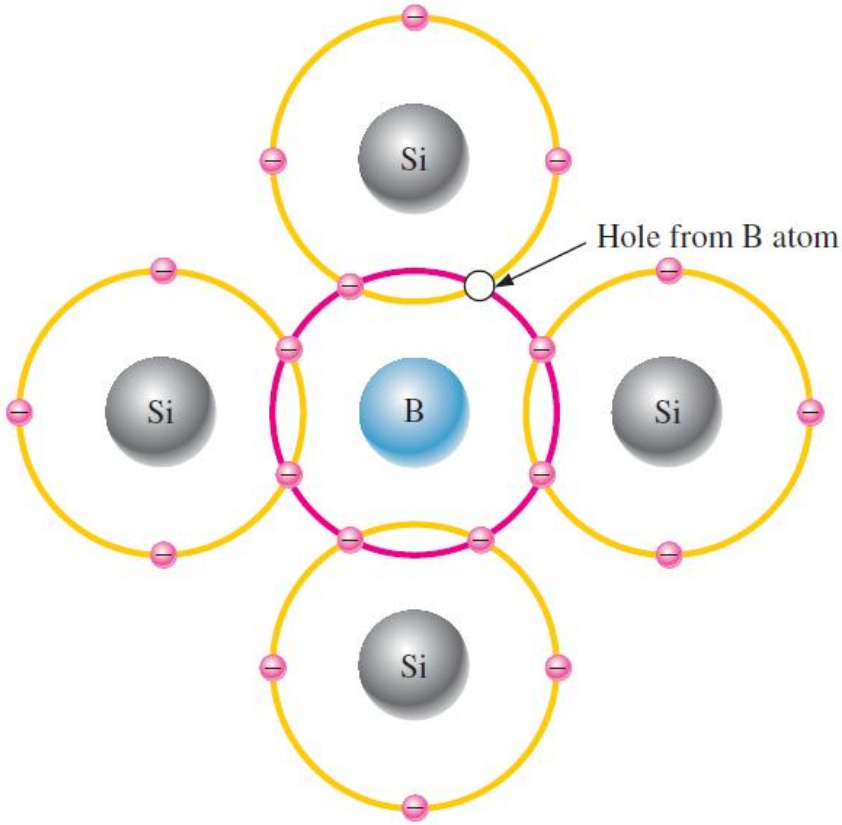


P-TYPE SEMICONDUCTORS

- *P-type semiconductors* are made by adding *3 valence electrons materials*, such as boron (B 硼), indium (In 銦), gallium (Ga 鎵), into the pure silicon or germanium.

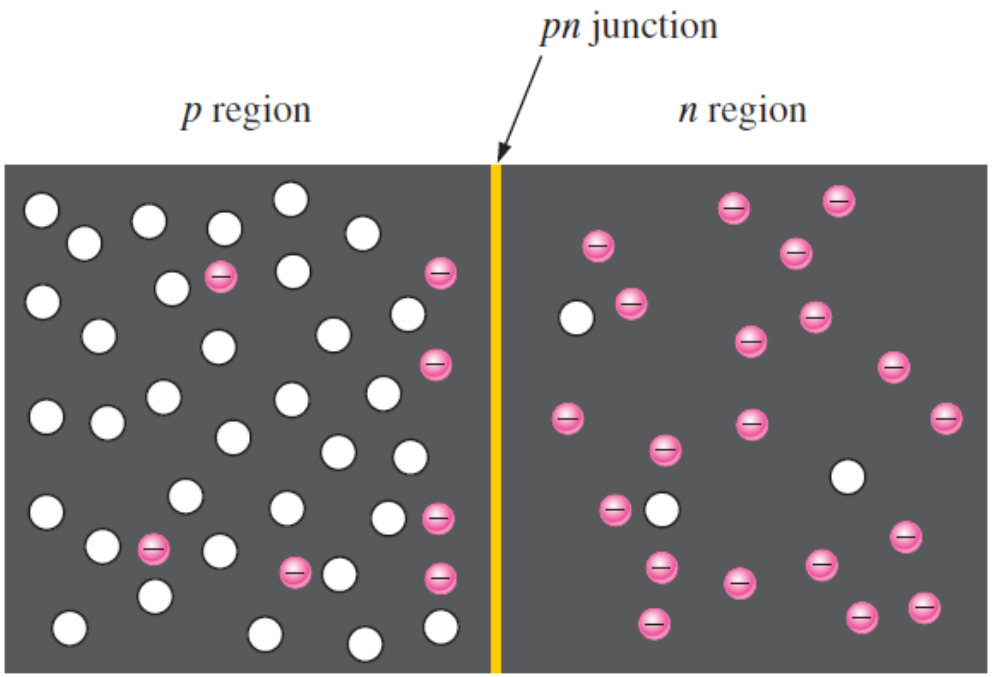
- *P-type semiconductors* have:
  - more holes (majority)
  - less electrons (minority)

- The "*P*" means the positive charge 正電荷 due to the holes.

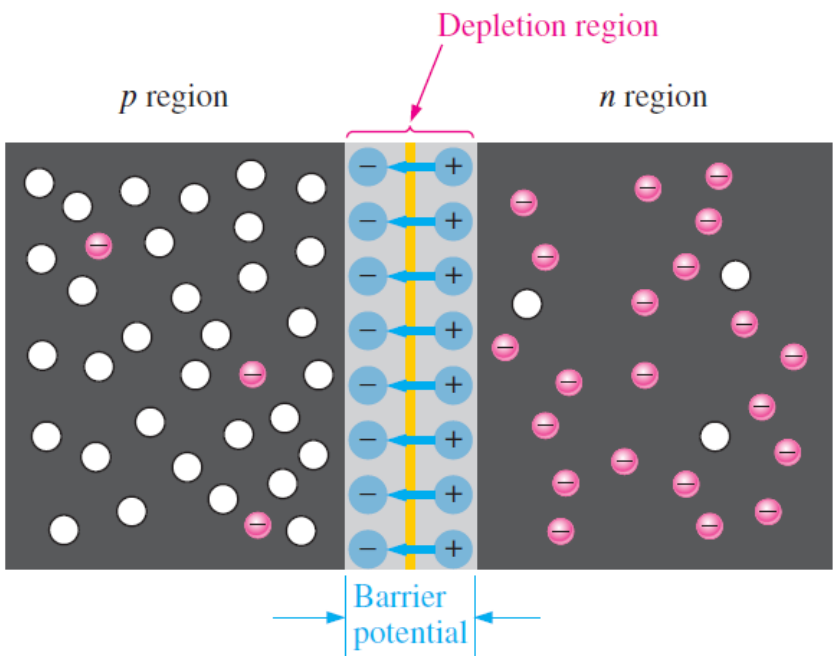


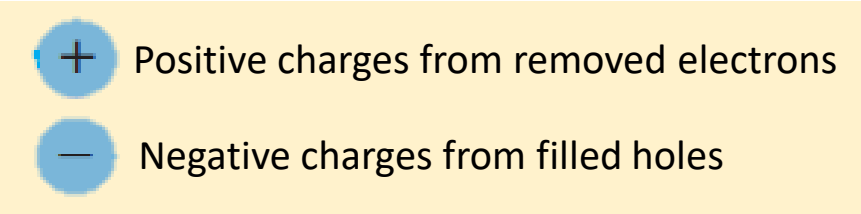
PN JUNCTION

- If a piece of pure silicon is doped so that one part is *n*-type and the other part is *p*-type, a *pn junction* (pn接面) *forms at the boundary between the two regions* and a diode 二極體 is created.



- When the *pn* junction is formed, the *n* region loses free electrons, creating **a layer of positive charges**. Meanwhile, the *p* region loses holes as the electrons and holes combine, creating **another layer of negative charges**.
- These *two layers of positive and negative charges* form the ***depletion region 空乏層***, creating an electric field as a ***barrier potential 位勢障礙***, which blocks the electron flow across the *pn* junction if no external energy is applied.
- In other words, a certain amount of voltage equal to the barrier potential must be applied across a *pn* junction before electrons will begin to flow across the junction.





Animation: https://youtu.be/7ukDKVHnac4?t=45

SUMMARY

1. Semiconductors require extra energy to be electrically conductive.
2. Most of semiconductive materials are made of either silicon (Si) or germanium (Ge).
3. To improve the conductivity, the pure semiconductive materials are doped, by adding impurities, creating p-type and n-type semiconductors.
4. There is a barrier potential at the *pn* junction, blocking the current flow if no additional energy is applied.